

SYNTHETIC CATEGORY THEORY

by D.-C. CISINSKI, B. CNOSSEN, K. NGUYEN and T. WALDE

STEFANO LUNEIA

LEARNING SEMINAR
UNIVERSITY OF OTTAWA

MOTIVATIONS AND GOALS

- CATEGORIES AS PRIMITIVE OBJECTS

$(\infty, 1)$ -CATEGORIES

SETS $\subset$ ANIMAE $\subset$ CATEGORIES

MOTIVATIONS AND GOALS

- CATEGORIES AS PRIMITIVE OBJECTS

$(\infty, 1)$ -CATEGORIES

SETS $\subset$ ANIMAE $\subset$ CATEGORIES
HoTT

MOTIVATIONS AND GOALS

• CATEGORIES AS PRIMITIVE OBJECTS

$(\infty, 1)$ -CATEGORIES

SETS $\subset$ ANIMAE $\subset$ CATEGORIES
HoTT

- CATEGORIES AND THEIR BASIC OPERATION AS PRIMITIVE
- STATE THEIR BEHAVIOUR THROUGH AXIOMS and UNIVERSAL PROPERTIES

MOTIVATIONS AND GOALS

• CATEGORIES AS PRIMITIVE OBJECTS

$(\infty, 1)$ -CATEGORIES

SETS $\subset$ ANIMAE $\subset$ CATEGORIES
HoTT

- CATEGORIES AND THEIR BASIC OPERATION AS PRIMITIVE
- STATE THEIR BEHAVIOUR THROUGH AXIOMS and UNIVERSAL PROPERTIES
- DEVELOP HIGHER CATEGORY THEORY IN ORDINARY MATHEMATICAL LANGUAGE
- NO COMBINATORICS OF A PARTICULAR MODEL
(AXIOMS PROVIDE A WAY TO HANDLE HIGHER COHEREN

• DIRECTED UNIVALENCE

In HoTT

$$(A =_u B) \cong \text{Equiv}(A, B)$$

• DIRECTED UNIVALENCE

In HoTT

$$(A =_u B) \simeq \text{Equiv}(A, B)$$

It is introduced a "CATEGORICAL UNIVERSE" Cat

$$\text{Hom}_{\text{Cat}}(C, D) \simeq \text{Map}(C, D)$$

• DIRECTED UNIVALENCE

In HoTT

$$(A =_u B) \simeq \text{Equiv}(A, B)$$

It is introduced a "CATEGORICAL UNIVERSE" Cat

$$\text{Hom}_{\text{Cat}}(C, D) \simeq \underbrace{\text{Map}(C, D)}_{\text{MAPPING ANIMA}}$$

• DIRECTED UNIVALENCE

In HoTT

$$(A =_u B) \simeq \text{Equiv}(A, B)$$

It is introduced a "CATEGORICAL UNIVERSE" Cat

$$\text{Hom}_{\text{Cat}}(C, D) \simeq \underbrace{\text{Map}(C, D)}_{\text{MAPPING ANIMA}}$$

ORDINARY UNIVALENCE REAPPEARS WHEN WE RESTRICT
TO INVERTIBLE STRUCTURES

"SYNTHETIC CATEGORY THEORY SHOULD EAT ITSELF"

- THE UNIVERSE $\mathbf{Cat}$ HAS SMALL SYNTHETIC CATS. AS OBJECTS
FUNCTORS AS MORPHISMS
- ITS INTERNAL LANGUAGE SHOULD SATISFY SYNTHETIC
CATEGORICAL AXIOMS
- DIRECTED UNIVALENCE $\rightsquigarrow$ INTERNAL DIAGRAM IN $\mathbf{Cat}$
CORRESPONDS TO FAMILIES
OF SYNTHETIC CATEGORIES

"SYNTHETIC CATEGORY THEORY SHOULD EAT ITSELF"

- THE UNIVERSE $\mathbf{Cat}$ HAS SMALL SYNTHETIC CATS. AS OBJECTS
FUNCTORS AS MORPHISMS
- ITS INTERNAL LANGUAGE SHOULD SATISFY SYNTHETIC
CATEGORICAL AXIOMS
- DIRECTED UNIVALENCE $\rightsquigarrow$ INTERNAL DIAGRAM IN $\mathbf{Cat}$
CORRESPONDS TO FAMILIES
OF SYNTHETIC CATEGORIES

THE FULL SELF-INTERPRETATION THEOREM
IS DEFERRED TO VOLUME II

AXIOM A.1

- Categories C, D, E ...
among them there are animae

AXIOM A.1

- Categories $C, D, E \dots$
among them there are

animae

∞ -GROUPOIDS

AXIOM A.1

- Categories $C, D, E \dots$
among them there are

anima

∞ -GROUPOIDS

- A anima, $x:A$ are called absolute objects

AXIOM A.1

- Categories $C, D, E \dots$
among them there are anima

∞ -GROUPOIDS

- A anima, $x:A$ are called absolute objects
- C, D categories, $\text{Map}(C, D)$ mapping anima

AXIOM A.1

- Categories $C, D, E \dots$
among them there are anima

∞ -GROUPOIDS

- A anima, $x:A$ are called absolute objects
- C, D categories, $\text{Map}(C, D)$ mapping anima
 $f: \text{Map}(C, D)$ called functor $f: C \rightarrow D$

AXIOM A.1

- Categories $C, D, E \dots$
among them there are anima

∞ -GROUPOIDS

- A anima, $x:A$ are called absolute objects
- C, D categories, $\text{Map}(C, D)$ mapping anima
 $f: \text{Map}(C, D)$ called functor $f: C \rightarrow D$
- $\text{id}_C: C \rightarrow C$
and given $f: C \rightarrow D, g: D \rightarrow E \rightsquigarrow g \circ f: C \rightarrow E$

- X anima and $x, y \in X$, $x \approx_x y$ anima of identifications

- X anima and $x, y: X$, $x \cong_x y$ anima of identifications

rules given by absolute objects

$$x: X, \text{ id}_x: x \cong x$$

$$\mu: x \cong y, \mu^{-1}: y \cong x$$

$$\mu: x \cong y, \nu: y \cong z, \nu\mu: x \cong z$$

- X anima and $x, y: X$, $x \cong_x y$ anima of identifications

rules given by absolute objects

$$x: X, \text{ id}_x: x \cong x$$

$$u: x \cong y, \text{ } u^{-1}: y \cong x$$

$$u: x \cong y, \text{ } v: y \cong z, \text{ } v u: x \cong z$$

- ACTION OF FUNCTORS

$F: A \rightarrow B$ functor between anima

$x: A$ there exists $Fx: B$

$u: x \cong_A y$ there exists $Fu: Fx \cong Fy$

- X anima and $x, y: X$, $x \cong_x y$ anima of identifications

rules given by absolute objects

$$x: X, \quad \text{id}_x: x \cong x$$

$$\mu: x \cong y, \quad \mu^{-1}: y \cong x$$

$$\mu: x \cong y, \quad \nu: y \cong z, \quad \nu\mu: x \cong z$$

- ACTION OF FUNCTORS

$F: A \rightarrow B$ functor between anima

$x: A$ there exists $Fx: B$

$\mu: x \cong_A y$ there exists $F\mu: Fx \cong Fy$

and $\text{id}_A x \cong x$ $(G \circ F)x \cong G(Fx)$

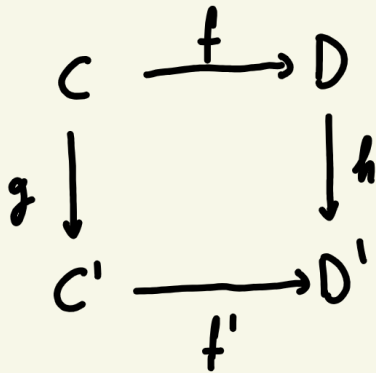
Def (NATURAL ISOMORPHISMS)

Given functors $f, f': C \rightarrow D$ a NATURAL ISOMORPHISM
is an identification $\alpha: f \cong f'$ in $\text{Map}(C, D)$

Def (NATURAL ISOMORPHISMS)

Given functors $f, f': C \rightarrow D$ a NATURAL ISOMORPHISM is an identification $\alpha: f \cong f'$ in $\text{Map}(C, D)$

(COMMUTATIVE SQUARE)



together with a NAT. ISO

$$\alpha: h \circ f \cong f' \circ g$$

AXIOMS FOR COHERENCES

- $f: C \rightarrow D$, $g: D \rightarrow E$, $h: E \rightarrow F$

$$\lambda_f: \text{id}_D \circ f \equiv f \qquad \rho_f: f \circ \text{id}_C \equiv f$$

$$\alpha_{f,g,h}: h \circ (g \circ f) \equiv (h \circ g) \circ f$$

AXIOMS FOR COHERENCES

• $f: C \rightarrow D$, $g: D \rightarrow E$, $h: E \rightarrow F$

$$\lambda_f: \text{id}_D \circ f \equiv f \qquad \rho_f: f \circ \text{id}_C \equiv f$$

$$\alpha_{f,g,h}: h \circ (g \circ f) \equiv (h \circ g) \circ f$$

+ UNITALITY and ASSOCIATIVITY for identifications

AXIOMS FOR COHERENCES

- $f: C \rightarrow D$, $g: D \rightarrow E$, $h: E \rightarrow F$

$$\lambda_f: \text{id}_D \circ f \equiv f \quad \rho_f: f \circ \text{id}_C \equiv f$$

$$\alpha_{f,g,h}: h \circ (g \circ f) \equiv (h \circ g) \circ f$$

+ UNITALITY and ASSOCIATIVITY for identifications

- HORIZONTAL and VERTICAL COMPOSITIONS for natural isomorphisms and identifications of nat. isomorphisms

consequence: INTERCHANGE RULE

- HORIZONTAL COMPOSITION IS COMPATIBLE WITH
UNITALITY and ASSOCIATIVITY ISOMORPHISMS

- HORIZONTAL COMPOSITION IS COMPATIBLE WITH
UNITALITY and ASSOCIATIVITY ISOMORPHISMS

- PENTAGON and TRIANGLE IDENTITIES

Def (TERMS)

Let Γ, C categories, a TERM OF C IN CONTEXT Γ
is an ABSOLUTE OBJECT of $\text{Map}(\Gamma, C)$

denoted as $\Gamma \vdash x : C$

Def (TERMS)

Let Γ, C categories, a TERM OF C IN CONTEXT Γ
is an ABSOLUTE OBJECT of $\text{Map}(\Gamma, C)$

denoted as

$$\Gamma \vdash x : C$$

$$x : \Gamma \longrightarrow C$$

Def (TERMS)

Let Γ, C categories, a TERM OF C IN CONTEXT Γ
is an ABSOLUTE OBJECT of $\text{Map}(\Gamma, C)$

denoted as $\Gamma \vdash x : C$ $x : \Gamma \rightarrow C$

Given $\sigma : \Delta \rightarrow \Gamma$ functor SUBSTITUTION OF x ALONG σ

is defined as

$$x[\sigma] := x \circ \sigma : \Delta \rightarrow C$$

Def (TERMS)

Let Γ, C categories, a TERM OF C IN CONTEXT Γ
is an ABSOLUTE OBJECT of $\text{Map}(\Gamma, C)$

denoted as $\Gamma \vdash x : C$ $x : \Gamma \rightarrow C$

Given $\sigma : \Delta \rightarrow \Gamma$ functor SUBSTITUTION OF x ALONG σ

is defined as

$$x[\sigma] := x \circ \sigma : \Delta \rightarrow C$$

$$\Delta \vdash x[\sigma] : C$$

Def (TERMS)

Let Γ, C categories, a TERM OF C IN CONTEXT Γ
is an ABSOLUTE OBJECT of $\text{Map}(\Gamma, C)$

denoted as $\Gamma \vdash x : C$ $x : \Gamma \rightarrow C$

Given $\sigma : \Delta \rightarrow \Gamma$ functor SUBSTITUTION OF x ALONG σ

is defined as

$$x[\sigma] := x \circ \sigma : \Delta \rightarrow C$$

$$\Delta \vdash x[\sigma] : C$$

Unitors and associators provides

$$x[\text{id}_\Gamma] \cong x$$

$$(x[\sigma])[\tau] \cong x[\sigma \circ \tau]$$

If Γ is an animal we call x an OBJECT of C in
CONTEXT Γ

The identity functor $\text{id}_C: C \rightarrow C$ is a term of C
in context C and we call it UNIVERSAL TERM OF C

If Γ is an animal we call x an OBJECT of C in
CONTEXT Γ

The identity functor $\text{id}_C : C \rightarrow C$ is a term of C
in context C and we call it UNIVERSAL TERM of C

WARNING NOTATION IS SUGGESTIVE BUT IT IS NOT
FORMAL TYPE THEORY

If Γ is an animal we call x an OBJECT of C in
CONTEXT Γ

The identity functor $\text{id}_C: C \rightarrow C$ is a term of C
in context C and we call it UNIVERSAL TERM of C

WARNING NOTATION IS SUGGESTIVE BUT IT IS NOT
FORMAL TYPE THEORY

law for substitution hold up to (SPECIFIED) ISOMORPHISMS

Writing $F(x) := F \circ x$ REQUIRES AN ALREADY CONSTRUCTED
FUNCTOR $F: C \rightarrow D$

Def (EQUIVALENCES)

A functor $f: C \rightarrow D$ is an EQUIVALENCE if
there exists $g: D \rightarrow C$ s.t.

$$\text{id}_C \cong gf$$
$$\text{id}_D \cong fg$$

Def (EQUIVALENCES)

A functor $f: C \rightarrow D$ is an EQUIVALENCE if
there exists $g: D \rightarrow C$ s.t. $\text{id}_C \cong gf$
 $\text{id}_D \cong fg$

AXIOM Any category equivalent to an anima is
an anima

Prop Let $f: \Delta \longrightarrow \Gamma$ functor. If the following operations have been constructed in every categorical context $\Leftrightarrow$ then f is an equivalence

i)

ii)

Prop Let $f: \Delta \longrightarrow \Gamma$ functor. If the following operations have been constructed in every categorical context $\textcircled{H}$ then f is an equivalence

i) Given $\textcircled{H} \vdash x, y: \Delta$ and $\textcircled{H} \vdash \tau: f(x) \cong f(y)$
we can construct $\textcircled{H} \vdash \sigma: x \cong y$

ii)

Prop Let $f: \Delta \rightarrow \Gamma$ functor. If the following operations have been constructed in every categorical context $\mathbb{H}$ then f is an equivalence

i) Given $\mathbb{H} \vdash x, y: \Delta$ and $\mathbb{H} \vdash \tau: f(x) \cong f(y)$
we can construct $\mathbb{H} \vdash \sigma: x \cong y$

ii) Given $\mathbb{H} \vdash \gamma: \Gamma$ we can construct
 $\mathbb{H} \vdash x: \Delta$ and $\mathbb{H} \vdash \tau: f(x) \cong \gamma$

PROOF

Apply ii) to the universal term of Γ $\Gamma \vdash \gamma: \Gamma$
 $\text{id}_\Gamma: \Gamma \longrightarrow \Gamma$

hence we get $g: \Gamma \longrightarrow \Delta$

$$\Gamma \vdash g: \Delta$$

and an identification $\tau: fg \cong \text{id}_\Gamma$ $\Gamma \vdash \tau: fg \cong \text{id}_\Gamma$

PROOF

Apply ii) to the universal term of Γ $\Gamma \vdash y : \Gamma$
 $\text{id}_\Gamma : \Gamma \longrightarrow \Gamma$

hence we get $g : \Gamma \longrightarrow \Delta$

$$\Gamma \vdash g : \Delta$$

and an identification $\tau : fg \cong \text{id}_\Gamma$ $\Gamma \vdash \tau : fg \cong \text{id}_\Gamma$

Since $\Delta \vdash gf$, $\text{id}_\Delta : \Delta$ and τ we have

$$\sigma : f \circ (g \circ f) \underset{\tau \circ f}{\cong} (f \circ g) \circ f \underset{\tau \circ f}{\cong} \text{id}_\Gamma \circ f \cong f \cong f \circ \text{id}_\Delta$$

PROOF

Apply ii) to the universal term of Γ $\Gamma \vdash y : \Gamma$
 $\text{id}_\Gamma : \Gamma \longrightarrow \Gamma$

hence we get $g : \Gamma \longrightarrow \Delta$

$$\Gamma \vdash g : \Delta$$

and an identification $\tau : fg \cong \text{id}_\Gamma$ $\Gamma \vdash \tau : fg \cong \text{id}_\Gamma$

Since $\Delta \vdash gf$, $\text{id}_\Delta : \Delta$ and τ , we have

$$\sigma : f \circ (g \circ f) \underset{\tau \circ f}{\cong} (f \circ g) \circ f \underset{\tau \circ f}{\cong} \text{id}_\Gamma \circ f \cong f \cong f \circ \text{id}_\Delta$$

apply i) in context Δ to gf , id_Δ and identification σ

to get $g \circ f \cong \text{id}_\Delta$

□

Def (SECTIONS / RETRACTIONS)

Let $f: C \rightarrow D$ functor

- SECTION OF f $s: D \rightarrow C$ with $fs \cong \text{id}_D$
- RETRACTION OF f $r: D \rightarrow C$ with $\text{id}_C \cong rf$

Def (SECTIONS / RETRACTIONS)

Let $f: C \rightarrow D$ functor

- SECTION OF f $s: D \rightarrow C$ with $fs \cong \text{id}_D$
- RETRACTION OF f $r: D \rightarrow C$ with $\text{id}_C \cong rf$

FACTS

- f equiv iff HAS SECTION s and RETRACTION r
and $s \cong r$
- Any two inverses of an equivalence are ISOMORPHIC
- EQUIVALENCES are CLOSED UNDER COMPOSITION
- EQUIVALENCES satisfy 2-OUT-OF-3 PROPERTY
- EQUIVALENCES satisfy 2-OUT-OF-6 PROPERTY

Def. (RETRACT)

A functor $f: C \rightarrow D$ is a RETRACT of $f': C' \rightarrow D'$ if there exist a commutative diagram

$$\begin{array}{ccccc} C & \xrightarrow{g} & C' & \xrightarrow{h} & C \\ f \downarrow & & \downarrow f' & & \downarrow f \\ D & \xrightarrow{k} & D' & \xrightarrow{j} & D \end{array}$$

with $\alpha: id_C \cong hg$, $\beta: id_D \cong jk$ s.t. composing

with α and β is isomorphic to $id_f: f \cong f$

Def. (RETRACT)

A functor $f: C \rightarrow D$ is a RETRACT of $f': C' \rightarrow D'$ if there exist a commutative diagram

$$\begin{array}{ccccc} C & \xrightarrow{g} & C' & \xrightarrow{h} & C \\ f \downarrow & & \downarrow f' & & \downarrow f \\ D & \xrightarrow{k} & D' & \xrightarrow{j} & D \end{array}$$

with $\alpha: id_C \cong hg$, $\beta: id_D \cong jk$ s.t. composing
with α and β is isomorphic to $id_f: f \cong f$

FACT RETRACTS OF EQUIVALENCES ARE EQUIVALENCES